UIDAI DATA HACKATHON 2026

AadhaarVista
Aadhaar Enrolment & Update Intelligence

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Abstract

This report presents the **AadhaarVista: Aadhaar Enrolment & Update Intelligence**, a data-driven analytical project aimed at understanding national-level patterns and trends in Aadhaar enrolment and update activities across India. Aadhaar forms the backbone of India's Digital Public Infrastructure (DPI), enabling citizens to securely access a wide range of public and private services. As Aadhaar coverage has reached a mature stage, the focus has gradually shifted from expanding enrolments to efficiently managing the identity lifecycle, with greater emphasis on demographic and biometric updates to ensure data accuracy and service continuity.

The analysis is carried out using **Exploratory Data Analysis (EDA)** and an **interactive Power BI dashboard**. EDA is used to analyse and describe the Aadhaar enrolment data by understanding its structure, data distributions, and relationships between key variables. This includes exploring enrolment patterns, examining correlations, and performing **time series trend analysis with short-term forecasting** to understand temporal variations in Aadhaar enrolment and update activity. These steps help uncover underlying enrolment and update behaviours present in the dataset. **Power BI** is used to compute, track, and visualize **key performance indicators (KPIs)** such as **total Aadhaar enrolments, adult population counts, biometric update cases, and demographic update cases**. The dashboard provides a consolidated, intuitive, and interpretable view of Aadhaar service activity, enabling effective comparison across regions and supporting data-driven monitoring and decision-making.

The outcome of AadhaarVista is a **decision-support analytics and visualization system** that transforms raw Aadhaar data into actionable insights. It enables UIDAI and policymakers to identify high-activity regions, understand update pressure trends, and support data-driven planning for improved service efficiency and infrastructure management. The project demonstrates how EDA combined with KPI-driven dashboards can enhance transparency, monitoring, and strategic decision-making within Aadhaar service operations

Problem Definition & Scope

2.1 Background and Context

Aadhaar serves as the backbone of India's Digital Public Infrastructure (DPI), enabling secure and reliable identity-based access to a wide range of government and private services. With Aadhaar adoption reaching near-universal coverage, the focus has gradually shifted from large-scale enrolment to the **efficient management of the Aadhaar identity lifecycle**. This includes continuous demographic and biometric updates required to maintain data accuracy and service reliability.

2.2 Problem Definition

Despite the availability of large volumes of Aadhaar enrolment and update data, extracting actionable insights from raw datasets remains challenging.
Key challenges addressed in this project include:

- Lack of consolidated analytical views to understand enrollment trends across states and districts.
- Limited visibility into age-group-wise enrolment behavior and update patterns.
- Challenges in monitoring update pressure caused by demographic and biometric updates.
- Absence of visual representations that can support data-driven monitoring and decision-making.

Without structured analysis and visualization, it becomes difficult to support data-driven decisions related to service optimization and infrastructure planning.

2.3 Project Scope

The scope of the project **AadhaarVista: Aadhaar Enrolment & Update Intelligence** is defined to address the above challenges through structured analysis and visualization. The project focuses on:

- Analyzing **anonymized Aadhaar enrolment and update data** provided by UIDAI.
- Performing **Exploratory Data Analysis (EDA)** to understand data distributions, correlations, and regional patterns
- Conducting **time series analysis with short-term forecasting** to observe temporal trends in Aadhaar enrolment.
- Designing an **interactive Power BI dashboard** to visualize key performance indicators (KPIs)

The analysis is limited to the **time period from 01 April 2025 to 31 December 2025** and is based solely on the Aadhaar enrolment and update dataset provided for the UIDAI Data Hackathon 2026.

Dataset Description

3.1 Dataset Source

The dataset used in this project is the **Aadhaar Enrolment and Update dataset provided by the Unique Identification Authority of India (UIDAI)** as part of the **UIDAI Data Hackathon 2026**. The dataset is fully anonymized and shared exclusively for analytical and visualization purposes in compliance with UIDAI data usage guidelines. No personally identifiable information (PII) is included or used in this study.

- **Time Period:** 01 April 2025 to 31 December 2025
- **Granularity:** State-level and District-level data

The dataset captures Aadhaar enrolment and update activity across different regions over the specified time period, enabling both regional and temporal analysis.

3.2 Dataset Components and Key Fields

Aadhaar Enrolment Dataset

- Folder: api_data_aadhar_enrolment
- Key Fields: date, state, district, pincode, age_0_5, age_5_17, age_18_greater

Aadhaar Biometric Update Dataset

- Folder: api_data_aadhar_biometric
- Key Fields: date, state, district, pincode, bio_age_5_17, bio_age_17_

Aadhaar Demographic Update Dataset

- Folder: api_data_aadhar_demographic
- Key Fields: date, state, district, pincode, demo_age_5_17, demo_age_17_

3.3 Data Integration and Usage

During the Exploratory Data Analysis (EDA) phase:

- CSV files within each folder are merged to reconstruct complete datasets
- Aggregations are performed at state, district, and monthly levels
- Derived metrics such as **total enrolments** and **update pressure indicators** are computed
  This integrated structure supports consistent analysis and dashboard-based KPI visualization.

Methodology

4.1 Data Cleaning & Preprocessing

The raw Aadhaar datasets provided for the UIDAI Data Hackathon 2026 were subjected to a structured data cleaning and preprocessing pipeline to ensure data quality, consistency, and analytical readiness. Since the dataset was distributed across multiple CSV files and folders due to its large size, the first step involved consolidating the data into a unified and usable format.

All CSV files within each dataset category (Aadhaar enrolment, biometric updates, and demographic updates) were **merged programmatically** to reconstruct complete datasets. Each file shared an identical schema, allowing seamless vertical concatenation without loss of information. This step ensured that the full data range was available for analysis.

```
[69]: # Enrolment dataset
enrolment = [
    "api_data_aadhar_enrolment_0_500000.csv",
    "api_data_aadhar_enrolment_500000_1000000.csv",
    "api_data_aadhar_enrolment_1000000_1006029.csv"
]

enrolment_df = pd.concat(
    [pd.read_csv(file) for file in enrolment],
    ignore_index = True
)

enrolment_df.head()
```

[69]:

	date	state	district	pincode	age_0_5	age_5_17	age_18_greater
0	02-03-2025	Meghalaya	East Khasi Hills	793121	11	61	37
1	09-03-2025	Karnataka	Bengaluru Urban	560043	14	33	39
2	09-03-2025	Uttar Pradesh	Kanpur Nagar	208001	29	82	12
3	09-03-2025	Uttar Pradesh	Aligarh	202133	62	29	15
4	09-03-2025	Karnataka	Bengaluru Urban	560016	14	16	21

The **date fields** were converted into a standard datetime format to support temporal operations such as sorting, grouping, and aggregation. Records were examined for missing or inconsistent values, and appropriate handling techniques were applied to maintain data integrity. As the dataset primarily contained aggregated counts, missing values were minimal and did not significantly impact analysis.

To enable trend-based analysis and reduce daily-level noise, the data was **aggregated from daily records to monthly summaries**. Monthly aggregation allowed clearer observation of enrolment and update trends over time and served as the basis for time series analysis and forecasting.

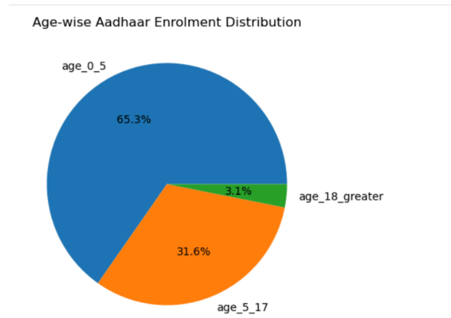
Additionally, **derived features** were created to support further analysis. These included monthly total enrolments computed from age-group-wise counts, as well as aggregated biometric and demographic update totals. These derived metrics were later used for correlation analysis, temporal modeling, and KPI computation in the Power BI dashboard.

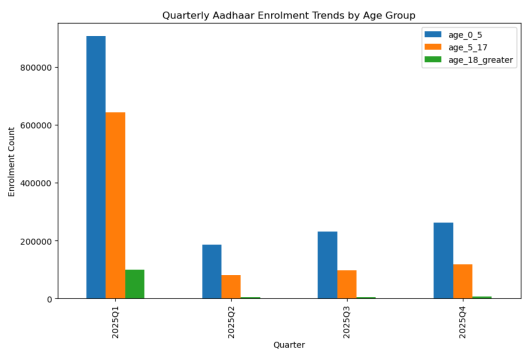
Through these preprocessing steps, the raw Aadhaar data was transformed into a clean, structured, and analysis-ready format, forming a reliable foundation for Exploratory Data Analysis and dashboard-based visualization.

4.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to analyze and describe the Aadhaar enrolment and update data, understand its underlying structure, and identify meaningful patterns and relationships within the dataset. This step played a crucial role in gaining insights before proceeding to visualization and dashboard development.

Initially, **data distribution analysis** was performed to understand the spread and magnitude of Aadhaar enrolments across different age groups, namely 0–5 years, 5–17 years, and 18 years and above. This helped in identifying the relative contribution of each age group to overall enrolment activity and understanding dominant enrolment categories.





To examine **temporal trends in Aadhaar enrolment**, quarterly aggregation was performed for each age group (0–5 years, 5–17 years, and 18+ years). The quarterly enrolment totals were visualized using a bar chart to compare enrolment patterns across different quarters. This analysis helps in understanding how enrolment activity for each age group changes over time and highlights periods of relatively higher or lower enrolment intensity.

To understand the relationship between Aadhaar enrolments across different age groups, a **correlation analysis** was performed for the age categories **0–5 years, 5–17 years, and 18+ years**. A correlation matrix was computed and visualized using a **heatmap**, which highlights the strength and direction of relationships between age-wise enrolment patterns. This analysis helps in identifying whether enrolment trends across age groups move independently or follow similar patterns over time.

Further, **state-wise and district-wise comparative analysis** was conducted to explore regional disparities in Aadhaar enrolment and update activity. Bar charts and heatmap visualizations were used to highlight variations across different geographical units and to identify high-activity regions.

Additionally, **heatmap-based visualizations** were employed to analyze regional and temporal patterns simultaneously. These visualizations provided an intuitive view of enrolment intensity across states and time periods, supporting effective identification of trends and concentration patterns.

Overall, the EDA phase enabled systematic exploration of the Aadhaar dataset, uncovering patterns related to age groups, regions, and enrolment behaviour. The insights obtained from this stage directly informed the design of KPIs and visual elements used in the Power BI dashboard.

4.3 Time Series Analysis & Forecasting

To analyze the temporal behaviour of Aadhaar enrolment activity, **time series analysis** was applied to the dataset using monthly aggregated data. Since the raw dataset contains daily records, the data was first aggregated at the **monthly level** to reduce short-term noise and enable clearer observation of trends over time.

Monthly enrolment totals, particularly for the **18+ age group**, were used to study enrolment patterns across the defined time period. **Trend visualization** techniques were employed to observe fluctuations, growth patterns, and overall movement in Aadhaar enrolment activity across months. These visualizations provided insight into periods of relatively higher or lower enrolment activity.

In addition to trend analysis, a **short-term forecasting approach using linear regression** was applied to the monthly aggregated data to understand the **directional movement** of enrolment trends. The forecasting was used only to observe near-term patterns based on historical data and was **not intended for long-term prediction or policy-level forecasting**. This ensured that the analysis remained exploratory and aligned with the scope of the dataset.

The time series analysis and forecasting complemented the Exploratory Data Analysis by adding a temporal perspective to enrolment behaviour. The insights derived from this stage helped in understanding how Aadhaar enrolment activity evolves over time and informed the design of time-based visual elements in the Power BI dashboard.

4.4 Dashboard Design & KPI

To translate analytical insights into an interpretable and decision-support format, an **interactive dashboard** was developed using **Microsoft Power BI**. Power BI was selected due to its capability to handle large datasets, support dynamic visualizations, and enable effective KPI monitoring through interactive filters and drill-down features.

The dashboard design is directly informed by insights obtained from **Exploratory Data Analysis (EDA)** and **time series analysis**. Visual elements were carefully selected to present both **aggregated metrics** and **regional comparisons** in a clear and intuitive manner. The dashboard enables users to explore Aadhaar enrolment and update activity across **states, districts, and time periods**, providing both high-level summaries and detailed breakdowns.

Key Performance Indicators (KPIs)

To provide a concise summary of Aadhaar service activity and facilitate quick assessment of enrolment and update pressure, the following KPIs are defined and displayed using **KPI cards**:

- **Total Aadhaar Enrolments:** Represents the overall volume of Aadhaar enrolments recorded during the selected time period, indicating the scale of enrolment operations.
- **Total Demographic Update Cases:** Captures the total number of demographic updates such as name, address, or date-of-birth modifications.

- **Total Biometric Update Cases:** Indicates the volume of biometric updates, reflecting identity re-verification requirements.
- **Update Pressure Ratio:** A derived indicator used to assess the relative intensity of update activity in comparison to enrolments.

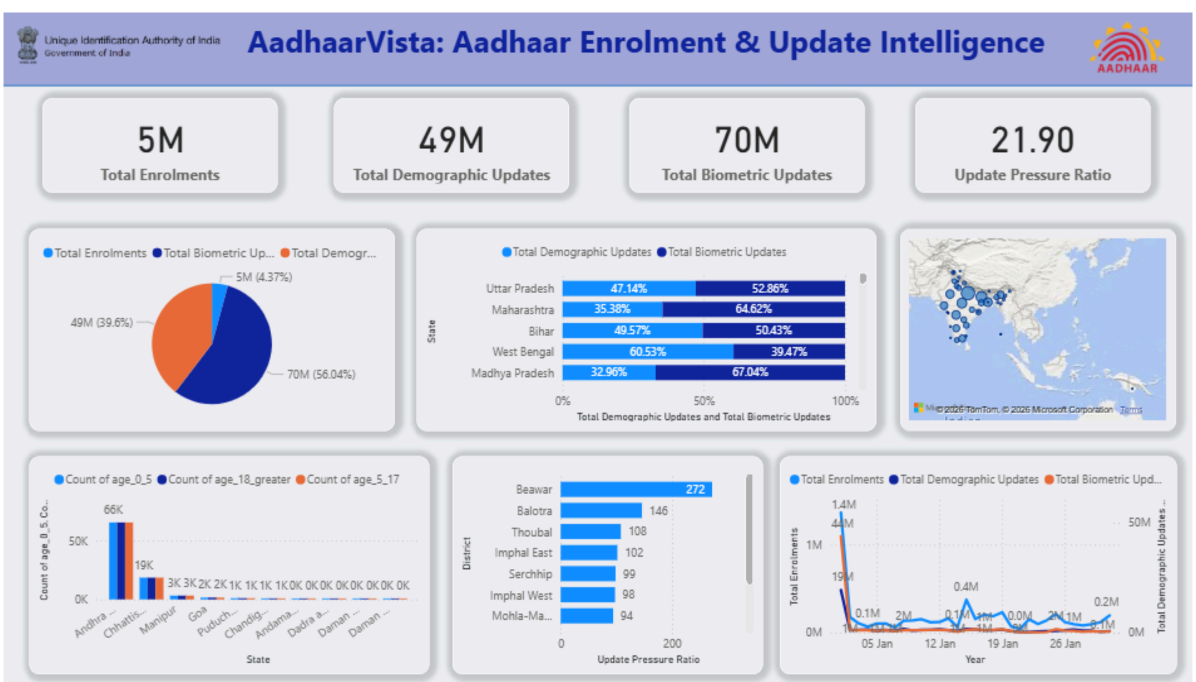
These KPIs provide instant visibility into Aadhaar service activity and form the entry point for further exploration within the dashboard.

Dashboard Visual Elements

The dashboard integrates multiple visual components to enhance interpretability and analytical depth:

- **Pie Chart:** Displays the proportional distribution of total enrolments, demographic updates, and biometric updates, offering a clear overview of Aadhaar service composition.
- **Stacked Bar Chart (State-wise):** Compares demographic and biometric updates across major states, highlighting regional differences in update behaviour.
- **Geographical Map Visualization:** Illustrates the spatial distribution of Aadhaar activity across India, enabling quick identification of high-activity regions.
- **Age-wise Bar Chart:** Shows enrolment distribution across age groups (0–5, 5–17, and 18+), supporting demographic analysis.
- **District-level Bar Chart:** Highlights districts with higher update pressure, aiding in understanding localized service demand.
- **Time Series Line Chart:** Displays trends in enrolments and updates over time, supporting temporal analysis.

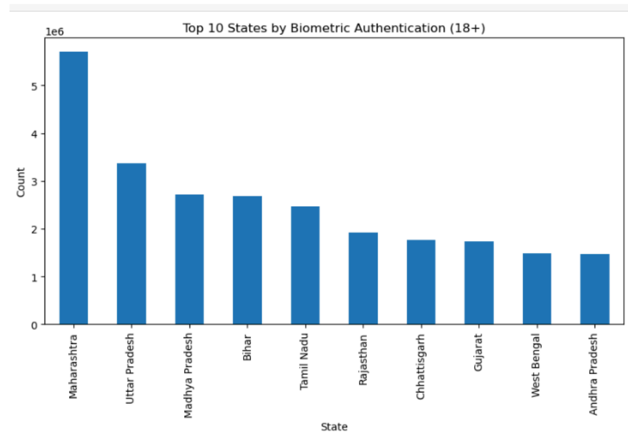
AadhaarVista Dashboard:



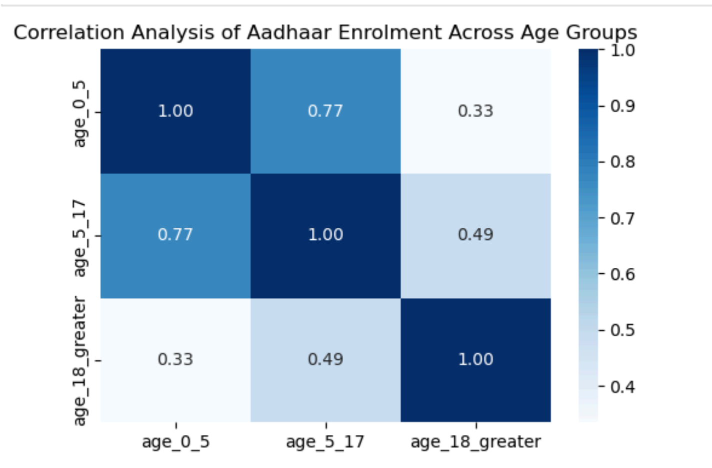
DATA ANALYSIS & VISUALISATION

5.1 Key Insights from EDA

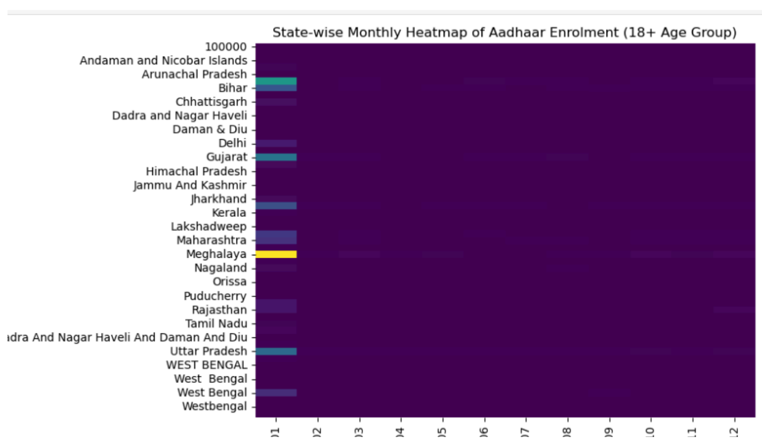
The Exploratory Data Analysis (EDA) provided meaningful insights into Aadhaar enrolment and update patterns across India. One of the key observations is that **children enrolments (under 5 age group) constitute the largest share of total Aadhaar enrolments**, indicating higher participation from the younger population compared to adult age groups.



Regional analysis reveals that **Aadhaar enrolment and biometric update activity varies significantly across states and districts**. Certain regions consistently exhibit higher enrolment and update volumes, while others record relatively lower activity. This variation highlights the uneven distribution of Aadhaar service demand across the country and points toward region-specific enrolment and update dynamics.



The relationship between enrolments across different age groups shows a **strong positive correlation**, indicating that enrolment activity across age categories tends to follow similar trends over time. This suggests that Aadhaar enrolment patterns are largely driven by broader systemic or administrative factors rather than isolated age-group-specific behaviour.

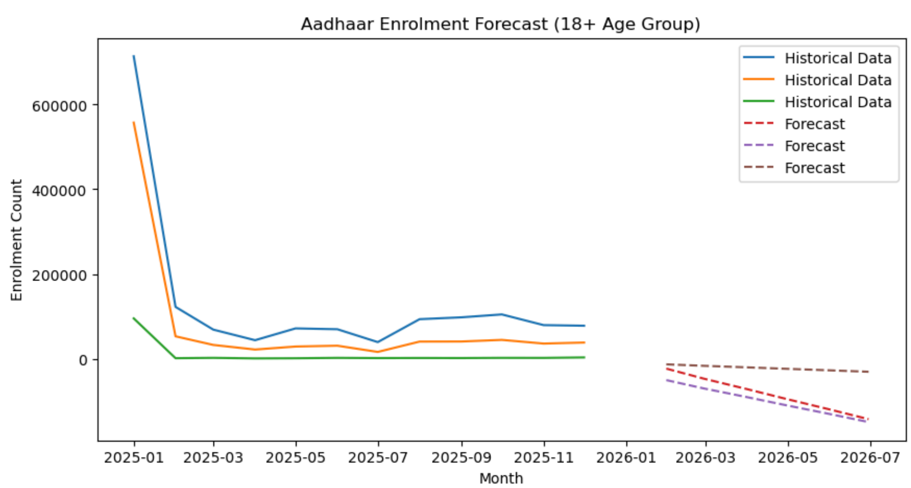


The state-wise monthly heatmap for the **18+ age group** provides a combined regional and temporal view of enrolment activity. The heatmap highlights variations in enrolment intensity across states and months, making it easier to identify states with consistently higher enrolment activity as well as periods of relatively increased or decreased enrolment.

The time series forecasting analysis applies a **linear regression - based approach** on monthly aggregated Aadhaar enrolment data for the **18+ age group** to understand short-term directional trends in enrolment activity. The historical time series shows an initial high enrolment volume followed by a period of stabilization with moderate fluctuations across subsequent months. This behaviour indicates that while enrolment activity experiences short-term variability, it largely follows a stable underlying trend over time rather than exhibiting abrupt or irregular changes. Linear regression is used to model this overall temporal pattern by capturing the general direction of movement across the historical data points.

The forecast generated through linear regression extends the observed trend into the near future, providing an indication of potential short-term movement in enrolment activity based on past behaviour. The projected values suggest a gradual continuation of the existing trend, reflecting stable enrolment dynamics rather than sharp increases or declines. This forecasting approach is intentionally limited in scope and is used for **exploratory trend interpretation**, not for precise prediction or long-term planning. The simplicity of the linear regression model ensures transparency and interpretability, making it suitable for high-level trend assessment within the constraints of the available dataset.

Overall, the time series forecasting complements the Exploratory Data Analysis by adding a temporal and predictive perspective to the study. It helps in understanding how Aadhaar enrolment activity for the adult population evolves over time and provides an indicative outlook that supports dashboard-based monitoring and short-term planning. The results reinforce the observation that Aadhaar enrolment trends remain relatively stable with controlled fluctuations, further validating the insights derived from descriptive and regional analyses.



Overall, the EDA helped uncover underlying patterns related to age groups, regions, and time, forming a strong analytical foundation for dashboard-based visualization and KPI interpretation.

Github Link:

[Purvijain1234/AadhaarVista]

5.2 Power BI Dashboard Analysis

The Power BI dashboard presents a consolidated view of Aadhaar enrolment and update activity using key metrics and visualizations derived from the dataset. The dashboard highlights the relative scale of enrolment and update operations and enables comparison across regions and time.

The KPI cards indicate that the dataset records approximately **5 million total Aadhaar enrolments**, alongside **49 million demographic update cases** and **70 million biometric update cases** during the analyzed period. This immediately reflects that **update activities significantly exceed fresh enrolments**, reinforcing the shift of Aadhaar operations from enrolment expansion to lifecycle maintenance.

Visual analysis further shows that biometric updates form the largest share of total Aadhaar activity, followed by demographic updates. This suggests a higher frequency of biometric re-verification requirements compared to demographic modifications. State-wise and district-wise visualizations indicate that Aadhaar activity is **regionally concentrated**, with certain states and districts consistently showing higher enrolment and update volumes. The geographical map visualization reinforces this observation by highlighting clusters of higher Aadhaar service activity across specific regions of India.

The time series visualization in the dashboard shows relatively stable enrollment trends with periodic fluctuations, supporting the findings from the EDA and forecasting analysis. Together, the dashboard provides an integrated and intuitive representation of Aadhaar service dynamics.

5.3 KPI Interpretation

The **Total Aadhaar Enrolments (5M)** KPI represents the overall volume of new enrolments recorded in the dataset and provides an estimate of enrolment scale during the selected time period. Compared to update volumes, this value indicates that Aadhaar enrolment has largely matured, with fewer new enrolments relative to update cases.

The **Biometric Update Cases (70M)** KPI highlights the significant operational load associated with biometric maintenance. This high volume suggests frequent biometric updates across the population, likely driven by factors such as age-related changes, authentication requirements, and periodic re-verification. From a future planning perspective, this indicates the need for sustained biometric update infrastructure and capacity.

The **Demographic Update Cases (49M)** KPI reflects substantial activity related to demographic modifications such as address or personal detail updates. This suggests continuous changes in demographic information over time, reinforcing the importance of accessible and efficient demographic update services.

The **Update Pressure Ratio (21.90)** serves as a critical indicator of operational intensity, showing that update activities occur at a much higher rate compared to new enrolments. A high update pressure ratio implies that Aadhaar service demand is increasingly driven by updates rather than enrolment, which has important implications for resource allocation, staffing, and service center planning.

Dashboard Link:

https://drive.google.com/drive/folders/1dwQGZ23tWfnTupScfnl05KJSIGKtJ5zc?usp=sharing

Results & Outcomes

The **AadhaarVista: Aadhaar Enrolment & Update Intelligence** project successfully transforms raw Aadhaar enrolment and update data into meaningful and interpretable insights through a structured analytical approach and interactive visualization. By combining **Exploratory Data Analysis (EDA)** with a **Power BI dashboard**, the project delivers clear outcomes that support effective monitoring and understanding of Aadhaar service activity.

A key outcome of the project is the **enhanced visibility of Aadhaar enrolment patterns** across age groups, regions, and time periods. The analysis reveals how enrolment activity differs across states and districts, enabling a structured and comparative understanding of regional enrolment behaviour beyond static tabular representations.

The project also facilitates **easy identification of high-activity regions**. Through state-wise and district-wise comparisons and heatmap-based visualizations, regions with consistently higher enrolment and update activity can be quickly recognized, allowing stakeholders to visually assess areas of concentrated Aadhaar service demand.

Another significant outcome is the **improved understanding of update pressure** arising from biometric and demographic updates. By analyzing update volumes alongside enrolment data and presenting them through KPIs, the project highlights the operational load associated with Aadhaar updates and how this load varies across regions and time.

The **Power BI dashboard** functions as an effective **decision-support tool**, presenting key metrics and trends in a consolidated and interactive format. KPI cards, comparative charts, and time-series visuals enable users to explore Aadhaar data dynamically and derive actionable insights without requiring technical expertise.

Overall, the project demonstrates how data analytics and visualization can enhance transparency, improve interpretability, and support data-driven monitoring of large-scale public identity systems. AadhaarVista provides a practical analytical framework that can assist stakeholders in understanding Aadhaar service dynamics and supporting informed planning for efficient service management.

Conclusion

This project demonstrates the effective use of **Exploratory Data Analysis (EDA)** and **Power BI visualization** to analyze and interpret Aadhaar enrollment and update data. Through a structured analytical workflow, **AadhaarVista** successfully converts raw, large-scale datasets into meaningful insights.

The combination of EDA and interactive dashboards enables clear understanding of enrolment patterns, regional variations, update pressure, and temporal trends. AadhaarVista serves as a practical and interpretable decision-support system that can assist stakeholders in monitoring Aadhaar service activity and supporting data-driven planning.

Overall, the project highlights the value of analytics and visualization in enhancing transparency and interpretability within large-scale public identity systems.